Toward an unrestricted classification of rank-four Grothendieck rings

Abstract

We study rank-four categorification without a multiplicity bound. Pivotal, spherical and pseudo-unitary existence are equivalent for rank-four fusion rings. We classify pivotal categorifiability for rings with a proper nontrivial fusion subring and for rings whose Frobenius–Perron character field has degree at most two. The remaining pivotal question concerns self-dual rings with no proper nontrivial fusion subring and character field of degree three or four; in the cubic case the rational formal codegree is three or four. The proofs combine uniform twist-moment estimates with explicitly bounded certificates. We also separate the Diophantine descriptions of rank-four fusion rings from the question of uniform completeness of categorification obstructions.

Contents

1 Scope and notation **2**

2 Uniform auxiliary obstructions **2**
2.1 Higher vanishing moments . . . 2
2.2 A determinant obstruction without enumerating sphericalizations . . . 4
2.3 Extensions of the character ring of the symmetric group . . . 5

3 Invertible objects and proper fusion subrings **6**
3.1 Unrestricted classification with at most two dimensions . . . 6
3.2 Removing the multiplicity bound in the non-self-dual pivotal case . . . 7
3.3 A central involution forced by two twist moments . . . 8
3.4 The unrestricted Fibonacci-subring case . . . 10
3.5 Proper fusion subrings and Galois reduction . . . 13

4 Dimension fields and Galois pseudo-unitarity **13**
4.1 A multiplicity bound for the quadratic character branch . . . 13
4.2 A uniform bound for the repeated-quadratic exception . . . 16
4.3 Products of two quadratic character fields . . . 19
4.4 The cubic branch reduces to rational codegree three or four . . . 20

5 A bounded induction difference in the cubic codegree-three branch **21**

6 The unrestricted classification problem **23**
6.1 The remaining unrestricted problem . . . 23
6.2 All fusion rings versus categorifiable fusion rings . . . 23
6.3 Diophantine presentations and finite census data . . . 24
6.4 What remains to be proved . . . 24

1 Scope and notation

The bounded study [11] applies exact categorification criteria to the rank-four fusion-ring census through multiplicity 128 and gives unitary models for the fifteen known categorifiable rings. The present paper asks which parts of that classification can be made independent of the multiplicity. The finite theorem is an input here, not a consequence of the unrestricted reductions: there is no circular use of the two papers.

We use the same basis convention $1, b_1, b_2, b_3$, $(N_i)_{jk} = N_{ij}^k$, Frobenius–Perron dimensions d_i, and $D = \sum_i d_i^2$. A proper nontrivial fusion subring has rank two or three. A character field is generated by the values of one character; it need not be the field generated by all formal codegrees. We distinguish these fields throughout. For the complete realization table and the fixed labels $F_{m,j}$ we refer to [11].

The following are used from that paper: the standard arithmetic inputs, Galois reduction of dimension characters, the low-induction Gram identity, the two-moment bound, and positivity of squared central dimensions. Their uses below are explicitly cited. The dimension and parameter bounds proved here make each subsequent finite computation exhaustive. The two supporting scripts for the unrestricted Fibonacci and quadratic-character arguments are supplied separately from the bounded classification verifier.

The conclusions concern existence of a categorification of a given based ring. Equality of pivotal, spherical and pseudo-unitary *existence* does not assert that every category is pivotal or every spherical realization is pseudo-unitary without Galois conjugation. Nor do the results below prove an unrestricted general or pivotal classification of every rank-four fusion ring.

2 Uniform auxiliary obstructions

2.1 Higher vanishing moments

We use the arbitrary-rank higher-indicator theorem and its twist-sector corollary from [11, Theorem 3.1 and Corollary A.2]. For $W_\zeta(V) = \mathrm{FPdim}(I(V)_\zeta)$ they state that

$$(r+1)W_\zeta(V) \le D\,\mathrm{FPdim}(V) \quad \text{if } \nu_1(V) = \cdots = \nu_r(V) = 0.$$

Here $\mathcal{C}$ is pseudo-unitary with its canonical spherical structure. The proof uses the nonnegative polynomial $|1 + z + \cdots + z^r|^2$ and the moment identity

$$\sum_Z [I(V) : Z]\,\mathrm{FPdim}(Z)\theta_Z^n = D\nu_n(V). \tag{1}$$

In particular, non-self-dual simples have vanishing first and second indicators and satisfy

$$3\,\mathrm{FPdim}(I_{\mathrm{low}}(X)) \le D\,\mathrm{FPdim}(X). \tag{2}$$

Remark 2.1. The proof also gives an explicit bound when the indicators do not vanish:

$$(r+1)^2 W_\zeta(V) \le (r+1)D\,\mathrm{FPdim}(V) + 2D\,\mathrm{Re}\sum_{j=1}^{r}(r+1-j)\zeta^{-j}\nu_j(V).$$

Only positivity of the canonical dimensions and the indicator formula are used; no $*$-structure is assumed. Positivity must not be dropped in a general spherical category. The polynomial is the unnormalised Fejér kernel.

Corollary 2.2 (Aggregating nontrivial simples)**.** *Let $V = \bigoplus_{i>0} a_i X_i$, with $a_i \in \mathbb{Z}_{\ge 0}$, and let L_V be a known contribution to its twist-one mass. Then*

$$2L_V \le D\,\mathrm{FPdim}(V), \qquad 9L_V \le D(3\,\mathrm{FPdim}(V) + 2\sum_{i>0} a_i).$$

Proof. The first indicator vanishes on every summand, while $\nu_2(V) = \sum a_i \nu_2(X_i) \le \sum a_i$. Apply the degree-one and degree-two Fejér polynomials. Equivalently, add the individual bounds. □

Remark 2.3 (Optimality and a convenient weaker form). Let $d \ge 1$, let $\varepsilon \in \{-1, 0, 1\}$, and prescribe total mass Dd, first moment zero and second moment $D\varepsilon$ for a positive measure on the unit circle. The bound in [11, Proposition 3.2] is optimal for these three moments. Indeed, put $T = Dd$ and choose θ with $\cos\theta = -(d+\varepsilon)/(2d)$. The measure with mass

$$\frac{T(d+\varepsilon)}{3d+\varepsilon} \text{ at } 1, \qquad \frac{Td}{3d+\varepsilon} \text{ at each of } e^{i\theta}, e^{-i\theta}$$

has total mass T, first moment zero, and second moment $D\varepsilon$, and it attains equality. If the two latter points coincide, their masses are added. This optimality statement does not assert that these points are roots of unity or that the measure is categorically realizable.

For calculations, the simpler Fejér polynomial $1 + z + z^2$ gives

$$9L_X \le D(3d + 2\varepsilon) \le D(3d + 2).$$

The optimized upper bound is smaller than this last upper bound by exactly $2D/[9(3d+1)]$. Both estimates apply without a unitary structure; pseudo-unitarity supplies the positive dimensions used in their proofs.

Lemma 2.4 (Uniform discriminant estimate)**.** *Let $n \ge 1$, and write $9n^2 + 4 = t^2 d$ with $t \ge 1$ and $d > 1$ squarefree. For the conductor f of $\mathbb{Q}(\sqrt{d})$, put $\lambda = (\varphi(f) - |\mu(f)|)/\sqrt{f/d}$ as in [11, Lemma 4.1]. Then*

$$\frac{\lambda}{\sqrt{d}} \ge \frac{8}{\sqrt{10}}.$$

Proof. The integer $9n^2 + 4$ lies strictly between $(3n)^2$ and $(3n+1)^2$, so $d > 1$. Every odd prime $p \mid d$ satisfies $p \ne 3$ and $(3n)^2 \equiv -4 \pmod p$. Therefore $p \equiv 1 \pmod 4$; in particular, $p \ge 5$. Reducing $t^2 d = 9n^2 + 4$ modulo three gives $d \equiv 1 \pmod 3$.

First suppose d is odd. Then $d \equiv 1 \pmod 4$, $f = d$, and $\lambda = \varphi(d) - 1$. Since $d \ne 1, 5$, it has a prime factor $p \ge 13$. Write $d = ph$. Every prime factor q of h satisfies $q \ge 5$, whence $(q-1)/\sqrt{q} \ge 1$ and $\varphi(h) \ge \sqrt{h}$. Consequently

$$\frac{\varphi(d) - 1}{\sqrt{d}} = \frac{(p-1)\varphi(h) - 1}{\sqrt{ph}} \ge \frac{(p-2)\varphi(h)}{\sqrt{ph}} \ge \frac{p-2}{\sqrt{p}} \ge \frac{11}{\sqrt{13}} > \frac{8}{\sqrt{10}}.$$

Here $(p-2)/\sqrt{p} = \sqrt{p} - 2/\sqrt{p}$ increases with $p > 0$.

If d is even, write $d = 2h$. Then h is odd, $h \ge 5$ because $d \equiv 1 \pmod 3$ excludes $d = 2$, and every prime factor of h is $1 \pmod 4$. Now $f = 8h$, $\mu(f) = 0$ and $\lambda = 2\varphi(h)$. Each factor $(p-1)/\sqrt{p}$ of $\varphi(h)/\sqrt{h}$ is at least $4/\sqrt{5} > 1$, and there is at least one such factor. Hence

$$\frac{\lambda}{\sqrt{d}} = \sqrt{2}\frac{\varphi(h)}{\sqrt{h}} \ge \frac{4\sqrt{2}}{\sqrt{5}} = \frac{8}{\sqrt{10}}.$$

□

Corollary 2.5. *For every $n \ge 2$, the quadratic family R_n defined in [11, Section 4] admits no pivotal categorification.*

Proof. The case $n = 2$ is the conductor-forty boundary obstruction of [11]. For $n > 2$, write $9n^2 + 4 = t^2 d$. Lemma 2.4 gives

$$nt\lambda \geq \frac{8n}{\sqrt{10}}\sqrt{9n^2+4} > 7n^2 + 4.$$

The strict comparison follows by squaring positive quantities:

$$\frac{64}{10}n^2(9n^2+4) - (7n^2+4)^2 = \frac{(43n^2+20)(n^2-4)}{5} > 0.$$

Thus the parameter condition in [11, Proposition 4.3] always holds. Pivotality is sphericality on a self-dual ring, so the spherical exclusion applies. □

2.2 A determinant obstruction without enumerating sphericalizations

The field and divisibility conditions can be combined before determining any signed multiplication table. This gives a general-categorification obstruction without a bounded signed-table search.

Proposition 2.6 (Signed determinant bound)**.** *Let R be a commutative self-dual based ring of rank r, with fusion matrices $N_0 = I, N_1, \dots, N_{r-1}$, and put*

$$H_i = \left(\sum_j N_j^2\right)_{ii}.$$

Suppose that the field generated by all formal codegrees of R has degree at least r. For each distinct irreducible factor P of the formal-codegree polynomial put $e_P = \deg P$ and $n_P = |P(0)|$, and define

$$B = \prod_p p^{\lceil r \max_P\{v_p(n_P)/e_P\}\rceil}.$$

If R admits a fusion categorification over $\mathbb{C}$, then

$$B \leq \prod_{i=0}^{r-1} H_i.$$

Proof. Form the signed quotient A of the sphericalization and let χ be its spherical dimension character. Write $L = \mathbb{Q}(\chi(b_i) : 0 \leq i < r)$. The codegree-field condition gives $[L : \mathbb{Q}] \geq r$. The character map $A_{\mathbb{Q}} \to L$ is surjective: its image is a finite-dimensional integral domain over $\mathbb{Q}$, hence a field. Since $\dim_{\mathbb{Q}} A_{\mathbb{Q}} = r$, this map is an isomorphism. In particular $[L : \mathbb{Q}] = r$.

Let T_i denote signed multiplication by b_i. These are real symmetric matrices, and the signed Casimir

$$Q_T = \sum_i T_i^2 = I + \sum_{i>0} T_i^2$$

is positive definite. Its value under χ is $\delta = \sum_i \chi(b_i)^2$, and consequently $\det Q_T = N_{L/\mathbb{Q}}(\delta)$. For every original formal codegree f, the ratio δ/f is an algebraic integer by [11, Proposition 2.5]. Taking norms in L gives

$$v_p(\det Q_T) \geq r\, v_p(n_P)/e_P \qquad (f \text{ a root of } P).$$

The determinant is a positive integer, so it is a multiple of B. On the other hand $|(T_j)_{ik}| \leq (N_j)_{ik}$ implies

$$(Q_T)_{ii} = \sum_{j,k} (T_j)_{ik}^2 \leq H_i.$$

Hadamard's determinant inequality therefore gives $B \leq \det Q_T \leq \prod_i (Q_T)_{ii} \leq \prod_i H_i$. □

Corollary 2.7. *Suppose a self-dual rank-four ring has two irreducible quadratic formal-codegree factors with distinct quadratic fields and norms a, b. If it is categorifiable over $\mathbb{C}$, then*

$$\operatorname{lcm}(a,b)^2 \le H_0 H_1 H_2 H_3.$$

Proof. The two quadratic fields generate a biquadratic field. Their norms give $B = \operatorname{lcm}(a,b)^2$ in Proposition 2.6. □

Remark 2.8 (Sufficiency for d-numbers). The normalized norm inequalities are not generally sufficient for algebraic integrality of a quotient. They are sufficient when both numerator u and denominator v are d-numbers. Enlarge $\mathbb{Q}(u,v)$ to a normal field L. The d-number property makes their valuations constant on all primes over a fixed rational prime p. If the common ramification index is e, those valuations are $e\,v_p(n_u)/[\mathbb{Q}(u):\mathbb{Q}]$ and $e\,v_p(n_v)/[\mathbb{Q}(v):\mathbb{Q}]$, respectively. The normalized inequalities therefore make u/v integral at every finite prime. Only the necessary direction is used in the finite-classification checker.

2.3 Extensions of the character ring of the symmetric group

Consider the rings with basis $1, g, r, x$ and products

$$g^2 = 1, \quad gr = r, \quad r^2 = 1 + g + r, \quad gx = x, \quad rx = 2x, \quad x^2 = 1 + g + 2r + \kappa x$$

for $\kappa = 6n$, $n \ge 0$. The cases $n = 0, 1, 2$ are respectively $F_{2,3}$, $F_{6,59}$ and $F_{12,260}$. None admits a pivotal categorification. Put $d = 3n + \sqrt{9n^2+6}$; the dimension vector is $(1, 1, 2, d)$. The formal codegrees are $D, D', 3, 2$, and the respective dimensions of the four summands of $I(1)$ are

$$1, \quad 1 + nd, \quad 4 + 2nd, \quad 6 + 3nd.$$

The corresponding low restriction matrix is forced to be

$$L = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & n & 2n & 3n \end{pmatrix}.$$

Here is an explicit justification despite the rational dependence of $1, 1, 2$. The first column is the central unit; all other first entries equal one. The irrational coefficients force the last row. The first row of LL^{T}, equal to that of the full Gram matrix, is $(4, 2, 3, 6n)$. The dimension-$1 + nd$ column consequently has middle entries zero. If a, b are the middle entries of the dimension-$4 + 2nd$ column, then $a + 2b = 3$ and $0 \le a \le 2$, forcing $(a, b) = (1, 1)$; the remaining column has middle entries $(1, 2)$.

The low mass in $I(g)$ is $10 + 5nd$, while $D = 12 + 6nd$. Thus $2\,\mathrm{FPdim}(I_{\mathrm{low}}(g)) - D = 8 + 4nd > 0$, contradicting the first-moment bound. The case $n = 0$ is also excluded without a pivotal assumption: its Frobenius–Perron dimension is the integer 12, so every categorification is pseudo-unitary [2, Proposition 8.24]. For every n, the characters with codegrees 2 and 3 vanish on x. The remaining two characters are the Frobenius–Perron character and its Galois conjugate. Since the ring is self-dual, [11, Lemma 2.2] therefore removes the pseudo-unitary hypothesis from the pivotal exclusion.

3 Invertible objects and proper fusion subrings

3.1 Unrestricted classification with at most two dimensions

The multiplicity bound can be removed entirely when the basis dimensions assume at most two distinct values. This statement concerns arbitrary complex categorification, and hence applies simultaneously to all five existence questions.

Theorem 3.1. *Let R be a rank-four fusion ring whose simple Frobenius–Perron dimensions have at most two distinct values. Then R admits a complex categorification if and only if it is one of the eleven rings in the following table. Every listed ring admits a unitary categorification.*

Number of invertibles	*Fusion ring or defining parameters*	*Number of rings*
4	$\mathbb{Z}[C_4]$, $\mathbb{Z}[C_2 \times C_2]$	2
3	$\mathrm{NG}(C_3, m)$, $\quad m \in \{0, 2, 3, 6\}$	4
2	$gX = Y$, $X = X^*$, $X^2 = 1 + aX + bY$, $(a, b) \in \{(1, 0), (1, 1), (2, 2)\}$	3
2	$S(1)$	1
2	$K_0(\mathrm{Rep}(D_{10}))$	1

Here D_{10} is the dihedral group of order ten, g is the nontrivial invertible in the third row, and $\mathrm{NG}(C_3, m)$ denotes the near-group ring with the three group elements and one object X satisfying $X^2 = \sum_{g \in C_3} g + mX$. The third-row rings are unchanged up to based-ring isomorphism when a and b are interchanged.

Proof. The pointed case is precisely the first row. Suppose henceforth that the noninvertible dimensions all equal $d > 1$, and let G be the group of invertible basis elements. Schopieray's theorem [7, Lemma 3.1] reduces general categorifiability to the existence of a pseudo-unitary categorification.

We first show that $|G| \geq 2$. Otherwise, choose noninvertible basis elements X, Y with $Y \neq X^*$. Their product has no unit summand, so $d^2 = kd$ for a nonnegative integer k and $d = k$ is an integer. On the other hand, XX^* has exactly one unit summand and no other invertible summand, giving $d^2 = 1 + \ell d$ for an integer ℓ. This forces $d \mid 1$, a contradiction.

If $|G| = 3$, its action fixes the unique noninvertible object. Frobenius reciprocity forces the near-group fusion rule in the table. Larson's classification [4, Theorem 1.1] gives precisely $m \in \{0, 2, 3, 6\}$ in the pseudo-unitary setting. The unitary realizations are recorded in Table 1 of [11].

It remains to consider $G = \{1, g\}$ and the two noninvertible objects X, Y. If g interchanges them, the duality either fixes both or interchanges them. In the self-dual case, Frobenius reciprocity gives

$$X^2 = Y^2 = 1 + aX + bY, \qquad XY = g + bX + aY.$$

The pseudo-unitary restriction [1, Theorem 2.4] gives, up to interchanging X and Y, the three nonpointed parameter pairs in the table. In the non-self-dual case, $X^2 = Y^2$ is fixed by duality, so its coefficients on X, Y agree. Thus $R = S(m)$ for some $m \geq 1$. [11, Theorem 3.4] leaves exactly $m = 1$. All four rings have unitary realizations as stated in the census table.

Finally suppose g fixes both noninvertibles. For $X \neq Y$, the product XY^* has no invertible summand: the multiplicity of $h \in G$ equals $[hY : X] = 0$. Hence $d^2 = kd$, so d is an integer. The product XX^* contains both $1, g$ once, giving $d^2 = 2 + \ell d$. Therefore $d = 2$. If $X^* = Y$, then XY is invariant under duality, so its noninvertible coefficients agree. Its dimension

equation would be $4 = 2 + 4c$, impossible for integral c. Thus both objects are self-dual. Write

$$X^2 = 1 + g + aX + bY, \quad XY = bX + cY, \quad Y^2 = 1 + g + cX + eY.$$

The dimension equations become $a + b = 1$, $b + c = 2$, and $c + e = 1$. Nonnegativity forces $(a, b, c, e) = (0, 1, 1, 0)$, which is the representation ring of D_{10}. This exhausts the cases. The known unitary realizations prove sufficiency. □

Remark 3.2 (What this leaves open). The theorem is independent of multiplicity, but its dimension hypothesis is essential to the argument. In particular, it does not settle arbitrary self-dual rank-four rings with three or four distinct simple dimensions. The unrestricted exclusion of the family R_n for all $n \geq 2$ supplies one further complete infinite family; it does not prove that these families exhaust the remaining rings.

3.2 Removing the multiplicity bound in the non-self-dual pivotal case

Lemma 3.3. *Every pivotal structure on a nonpointed rank-four fusion category with a non-self-dual simple object is spherical.*

Proof. Write the simple objects as $1, X, Y, Z$, with $X^* = Z$ and $Y^* = Y$. The slope operator of a pivotal fusion category is the monoidal natural endomorphism of the identity whose scalar on a simple object V is $\mathrm{sl}(V) = \dim_l(V)/\dim_r(V)$; see [8, Section 4C]. Consequently $N_{UV}^W > 0$ implies $\mathrm{sl}(W) = \mathrm{sl}(U)\,\mathrm{sl}(V)$. The slopes of $1, Y$ are one, and those of X, Z are s, s^{-1}. We prove $s = 1$.

Use the general rank-four non-self-dual multiplication table

$$\begin{aligned} X^2 &= pX + lY + cZ, & XY &= qX + kY + lZ, \\ Y^2 &= 1 + kX + eY + kZ, & XZ &= 1 + pX + qY + pZ, \\ Z^2 &= cX + lY + pZ, & YZ &= lX + kY + qZ, \end{aligned}$$

where c, e, k, l, p, q are nonnegative integers. Associativity gives

$$kl + lc = lp + kq, \qquad kp + le + kc = 2lq + k^2,$$

$$l^2 + c^2 = 1 + q^2 + p^2, \qquad l^2 + k^2 + q^2 = 1 + 2pk + qe.$$

If $s \neq 1$, the occurrences of X in X^2 and of Y in XY force $p = k = 0$. The first associativity equation then gives $lc = 0$. If $l = 0$, the third equation gives $(c - q)(c + q) = 1$, hence $c = 1, q = 0$, contrary to the fourth equation. Therefore $l > 0$ and $c = 0$. The third equation now gives $(l - q)(l + q) = 1$, hence $l = 1, q = 0$; the second then gives $e = 0$. The resulting table is the pointed ring of $\mathbb{Z}/4\mathbb{Z}$, contrary to the hypothesis. Thus every slope is one, which is precisely sphericity. □

Proposition 3.4. *A rank-four non-self-dual fusion ring admits a pivotal categorification if and only if it admits a pseudo-unitary categorification.*

Proof. Only the forward implication requires proof. If the ring is pointed, it admits a unitary model. Otherwise the preceding lemma makes any pivotal realization spherical.

Let R be the ring. Its four complex characters are permuted by the involution $\chi \mapsto (b \mapsto \chi(b^*))$. On $R \otimes_{\mathbb{Z}} \mathbb{C}$ the involution has trace two, since it fixes the two basis elements $1, Y$ and exchanges X, Z. On the primitive idempotent basis its trace is the number of fixed characters. Hence precisely two characters are fixed by duality. These are exactly the real-valued characters, because characters of a commutative fusion ring satisfy $\chi(b^*) = \overline{\chi(b)}$.

One of these real characters is the Frobenius–Perron character d. If all its values are rational, they are integers, and every fusion categorification is weakly integral and therefore pseudo-unitary. Otherwise the Galois orbit of d has at least two members. It consists entirely of characters fixed by duality, since $d(b) = d(b^*)$ is preserved by field automorphisms. Thus it consists of both real-valued characters. The spherical dimension character is real-valued, so is Galois conjugate to d. Conjugating the category and its spherical structure gives positive Frobenius–Perron dimensions, hence a pseudo-unitary realization. □

Corollary 3.5. *With no multiplicity restriction, the rank-four non-self-dual fusion rings admitting pivotal categorifications are exactly the six rings*

$$\mathbb{Z}[\mathbb{Z}/4\mathbb{Z}], \quad \mathrm{NG}(\mathbb{Z}/3\mathbb{Z}, m)\ (m = 0, 2, 3, 6), \quad S(1).$$

All six admit unitary realizations.

Proof. The proposition reduces the classification to the pseudo-unitary case. Larson's classification gives these six rings and the additional candidate $S(2)$. The uniform exclusion of $S(m)$ for $m \geq 2$ removes that candidate. The known unitary models are the ones already recorded in the realization table: the cyclic group, Tambara–Yamagami, $\mathrm{Rep}(A_4)$, Izumi–Xu, the $m = 6$ near-group model, and the non-self-dual quadratic model. □

Corollary 3.6. *Every pivotal structure on a nonpointed rank-four fusion category is spherical. Consequently, pivotal and spherical categorifiability are equivalent for all rank-four fusion rings, without a multiplicity bound.*

Proof. In the self-dual case, left and right dimensions agree on each simple object. The non-self-dual case is Lemma 3.3. Every pointed rank-four ring has a unitary realization, which handles the remaining existence statement. □

3.3 A central involution forced by two twist moments

Consider a self-dual rank-four ring with exactly two invertible basis objects $1, g$, where g fixes the other two objects X, Y. Its products are

$$\begin{aligned} g^2 = 1, \qquad gX = X, \qquad gY = Y, \\ X^2 = 1 + g + aX + bY, \qquad XY = bX + cY, \\ Y^2 = 1 + g + cX + eY, \qquad 2 + ac + be = b^2 + c^2. \end{aligned} \tag{3}$$

All parameters are nonnegative integers.

Lemma 3.7. *A ring of the form (3) admits a pivotal categorification if and only if it admits a pseudo-unitary categorification.*

Proof. The rational character $(1, -1, 0, 0)$ splits off a copy of $\mathbb{Q}$ from $R \otimes \mathbb{Q}$. Its complementary algebra has dimension three. Consequently the degree k of the Frobenius–Perron character field is 1, 2, or 3. A pivotal categorification is spherical because every simple is self-dual; its dimension character cannot be $(1, -1, 0, 0)$.

If $k = 1$, its Frobenius–Perron dimension is an integer, and the category is pseudo-unitary. If $k = 3$, all three remaining characters form one Galois orbit, so [11, Lemma 2.2] applies. Suppose $k = 2$ and write $D = \mathrm{FPdim}(R)$. If D is rational, it is an integer and the same weak-integrality argument applies. Otherwise D generates the quadratic character field. Besides the two conjugate Frobenius–Perron characters, the only remaining character is rational. It cannot be the spherical dimension character, since the spherical dimension field must contain every formal codegree, including the nonrational number D. Thus the dimension character is Galois conjugate to Frobenius–Perron, which proves the assertion. □

Proposition 3.8. *Suppose $b, c > 0$ and that (3) admits a pseudo-unitary categorification $\mathcal{C}$. Then the object g lifts to an invertible object of order two and twist one in $\mathcal{Z}(\mathcal{C})$. Moreover $\mathcal{C}$ is a C_2-equivariantization of a rank-five fusion category on which the nontrivial group element fixes only the tensor unit.*

Proof. Use the canonical spherical structure and put $D = \mathrm{FPdim}(\mathcal{C})$. The character $(1, -1, 0, 0)$ has formal codegree 2. Thus

$$I(1) = 1 \oplus A \oplus B \oplus C, \qquad \dim(B) = D/2, \qquad \dim(A) + \dim(C) = D/2 - 1.$$

All four summands have twist one. The coefficient of g in $FI(1)$ is two, while $\dim \mathrm{End}(I(g)) = 4$, as follows directly from $FI(g) = \sum_U U \otimes g \otimes U^*$.

The first twist moment gives that the dimension of the low part of $I(g)$ is at most $D/2$. If g occurred in $F(B)$, it would already contribute $D/2$, and the other required occurrence of g would make this bound strictly fail. Hence $[F(B) : g] = 0$. The two occurrences of g are therefore in $F(A)$ and $F(C)$. They cannot both be in one column: that multiplicity two would exhaust $\dim \mathrm{End}(I(g)) = 4$, so $I(g)$ would have only twist-one constituents, contrary to its vanishing first twist trace. Thus both multiplicities are one. The low dimension is

$$L = D/2 - 1.$$

The residual value of $\dim \mathrm{End}(I(g))$ is two, so there are exactly two further simple summands, each with multiplicity one. Denote their positive dimensions by α, β and twists by u, v. The zeroth, first and second twist moments give

$$\alpha + \beta = D/2 + 1, \qquad \alpha u + \beta v = -(D/2 - 1), \qquad \alpha u^2 + \beta v^2 = D\nu_2(g) - (D/2 - 1).$$

If $\nu_2(g) = -1$, the last right-hand side has absolute value $3D/2 - 1 > D/2 + 1$, since $D > 2$, violating the triangle inequality. Therefore $\nu_2(g) = 1$. Equality in the triangle inequality for the last identity forces $u^2 = v^2 = 1$. The first identity involving twists then forces, after interchanging the two summands,

$$(\alpha, u) = (1, 1), \qquad (\beta, v) = (D/2, -1).$$

Let z be the central simple of dimension one. Its restriction contains g, and consequently $F(z) = g$.

There is no nontrivial central invertible whose underlying object is the unit. Such an object would give a tensor automorphism of the identity of $\mathcal{C}$, with scalars $\lambda_g, \lambda_X, \lambda_Y$. The relations $gX = X$, $Y \subset X^2$, and $X \subset Y^2$, together with self-duality, force respectively $\lambda_g = 1$, $\lambda_Y = \lambda_X^2 = 1$, and $\lambda_X = \lambda_Y^2 = 1$. Hence $z^2 = 1$. The order-two invertible z has dimension and twist one, so it generates a Tannakian subcategory $\mathrm{Rep}(C_2)$ of the center, embedded fully faithfully into $\mathcal{C}$ by the forgetful functor.

De-equivariantization gives a fusion category $\mathcal{D}$ with a C_2-action and $\mathcal{C} \simeq \mathcal{D}^{C_2}$. For clarity, the orbit description of equivariantization determines its simple objects without a dimension estimate. A fixed simple of $\mathcal{D}$ gives two simples of $\mathcal{C}$, interchanged by tensoring with the sign representation g, because $H^2(C_2, \mathbb{C}^\times) = 0$. A two-element orbit gives one simple fixed by g. In $\mathcal{C}$, the unit and g form the sole two-element orbit under tensoring with g; X and Y are fixed. Thus the action on $\mathrm{Irr}(\mathcal{D})$ has exactly one fixed simple, the unit, and two orbits of size two. In particular $\mathcal{D}$ has rank five. □

Theorem 3.9. *Among rings of the form (3), the only one admitting a pivotal categorification is $K_0(\mathrm{Rep}(D_{10}))$. There is no multiplicity restriction.*

Proof. By Lemma 3.7 we can assume pseudo-unitarity. First suppose $b, c > 0$ and form $\mathcal{D}$ as in Proposition 3.8. The dimensions of its simples are

$$1, u, u, v, v, \qquad u = \mathrm{FPdim}(X)/2, \quad v = \mathrm{FPdim}(Y)/2.$$

They are integers, by the fixed-point-free automorphism theorem [9, Theorem 2.3]. There is also a short rank-five verification of this integrality. The complexified fusion algebra is either $\mathbb{C}^5$ or $\mathbb{C} \oplus M_2(\mathbb{C})$. In the second case its sole one-dimensional character, Frobenius–Perron dimension, is Galois fixed. In the first case, the involution has trace one in the based basis and therefore fixes exactly one primitive idempotent, hence exactly one character. Every Galois conjugate of Frobenius–Perron is fixed by the involution, so it must be that same character. Its values are rational algebraic integers in either case.

An integral fusion ring with at most three distinct basis dimensions has a nontrivial pointed subring [9, Lemma 4.2]. The pointed part of $K_0(\mathcal{D})$ is invariant under the involution, so it has rank three or five. If its rank is three, the two remaining objects have a common integer dimension $v > 1$ and are fixed by the three invertibles. Their product with their dual therefore gives $v^2 = 3 + kv$ for an integer $k \geq 0$, forcing $v = 3$. After exchanging X, Y if necessary, their dimensions in $\mathcal{C}$ would then be $2, 6$. The equation for X^2 would read $4 = 2 + 2a + 6b$, contradicting $b > 0$. If its pointed rank is five, the dimensions of $\mathcal{C}$ are $(1, 1, 2, 2)$. The dimension equations in (3) force $(a, b, c, e) = (0, 1, 1, 0)$, the representation ring of D_{10}.

It remains to treat a zero among b, c. By exchanging X, Y, assume $b = 0$. Associativity reads $c(c - a) = 2$, so $c = 2$, $a = 1$. Thus the ring is an extension of $K_0(\mathrm{Rep}(S_3))$ from Section 2.3, with arbitrary coefficient $\kappa = e \geq 0$. Put $d = (\kappa + \sqrt{\kappa^2 + 24})/2$ and $D = 12 + \kappa d$. For $\kappa \notin \{0, 1, 5\}$, this D is quadratic, with minimal polynomial

$$t^2 - (\kappa^2 + 24)t + 6(\kappa^2 + 24).$$

The d-number criterion implies $6 \mid \kappa^2$, hence $6 \mid \kappa$; these parameters, including $\kappa = 0$, are excluded by Section 2.3. The only positive κ for which $\kappa^2 + 24$ is a square are 1 and 5, as follows by factoring $(s - \kappa)(s + \kappa) = 24$. For the two remaining parameters $\kappa \in \{1, 5\}$, the bounded-census verifier matches the complete tensors and checks their nonexistence certificates; see [11, Theorem 1.1]. The finite theorem is independent of the present unrestricted argument. This completes the proof. □

Corollary 3.10. *A rank-four fusion ring with a nontrivial invertible basis element admits a pivotal categorification if and only if it is one of the eleven rings in Theorem 3.1. These all admit unitary categorifications.*

Proof. For a non-self-dual ring with a nontrivial invertible, its noninvertibles are either a single object or a dual pair, and therefore have equal Frobenius–Perron dimensions. Theorem 3.1 applies. In the self-dual case, four invertibles give a pointed ring; three invertibles cannot occur because a group of order three is not self-dual. With two invertibles, their action either interchanges the two remaining objects, which have equal dimensions and are covered by Theorem 3.1, or fixes both, which is exactly Theorem 3.9. □

3.4 The unrestricted Fibonacci-subring case

Theorem 3.11. *A rank-four fusion ring containing the Fibonacci fusion ring admits a pivotal categorification over $\mathbb{C}$ if and only if it is one of*

$$K_0(\mathrm{Fib} \boxtimes \mathrm{Vec}(C_2)), \qquad K_0(\mathrm{Fib} \boxtimes \mathrm{Fib}), \qquad K_0(\mathrm{Ad}(E_8)).$$

All three admit unitary categorifications. No multiplicity bound is assumed.

Write $f^2 = 1+f$. On the two remaining basis elements, multiplication by f is a symmetric nonnegative integral matrix M satisfying $M^2 = I + M$. Its eigenvalues are $(1 \pm \sqrt{5})/2$, so, after reordering, $M = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Duality commutes with M and cannot exchange its two basis elements. Consequently the ring is self-dual and has the following form:

$$\begin{aligned} f^2 &= 1+f, \qquad fX = Y, \qquad fY = X+Y, \\ X^2 &= 1 + aX + bY, \qquad XY = f + bX + (a+b)Y, \\ Y^2 &= 1 + f + (a+b)X + (a+2b)Y, \qquad a, b \in \mathbb{Z}_{\geq 0}. \end{aligned} \tag{4}$$

Here rank-four fusion rings are commutative, and the displayed formulas follow from Frobenius reciprocity and associativity. This two-parameter family is also identified in [10, Proposition 3.3]. Put

$$\phi = \frac{1+\sqrt{5}}{2}, \quad s = a + b\phi, \quad x = \frac{s + \sqrt{s^2+4}}{2}, \quad H = \phi^2, \quad K = \phi + 2.$$

The dimension row is $(1, \phi, x, \phi x)$, and $D = K(1+x^2)$.

Lemma 3.12. *If $b > 0$, the rational fusion algebra in (4) is the quartic field $L = \mathbb{Q}(\phi, x)$. Pivotal categorifiability is equivalent to pseudo-unitary categorifiability. A necessary condition for any complex categorification is*

$$N = u^2 + uv - v^2 = r^2 \text{ or } 5r^2, \quad u = a^2 + b^2 + 4, \quad v = 2ab + b^2, \quad r \in \mathbb{Z}_{>0}. \tag{5}$$

Proof. Suppose $x \in \mathbb{Q}(\phi)$. Both x and $x^{-1} = x - s$ are algebraic integers, so $\varepsilon = \mathrm{Norm}(x) = \pm 1$. Writing x' for its quadratic conjugate, if $\varepsilon = -1$ then $s = x + x'$ is rational, contrary to $b > 0$. If $\varepsilon = 1$, then $s = x - x'$ has trace zero, contrary to $\mathrm{Tr}(s) = 2a + b > 0$. Thus $[L : \mathbb{Q}] = 4$. Evaluation at the Frobenius–Perron character identifies the four-dimensional rational fusion algebra with L; all its characters form one Galois orbit. Self-duality makes every pivotal structure spherical, and [11, Lemma 2.2] gives a pseudo-unitary Galois conjugate.

Cyclotomicity of Frobenius–Perron dimensions [2] implies that $L/\mathbb{Q}$ is abelian, in particular normal. Put $\Delta = s^2 + 4 = u + v\phi$. The conjugate of Δ is $\Delta' = (a + b(1-\phi))^2 + 4$, so $N = \Delta\Delta' > 0$. Normality of $\mathbb{Q}(\phi)(\sqrt{\Delta})$ implies $\sqrt{\Delta'} \in L$. Since Δ' is not a square in $\mathbb{Q}(\phi)$, comparison of coefficients in this quadratic extension shows that Δ'/Δ is a square in $\mathbb{Q}(\phi)$. Hence N is a square in $\mathbb{Q}(\phi)$. A positive rational integer is a square in this quadratic field precisely when it is an integral square or five times an integral square, proving (5). □

We next determine the twist-one constituents of $I(1)$ without searching over integral restriction matrices. Write $h = \sqrt{N} > 0$ and $s' = a + b(1-\phi)$. The four formal codegrees are the four conjugates of D. The involution fixing ϕ and sending x to $-x^{-1}$ gives the ratio x^2. For the other two conjugates, direct use of $x^2 = 1 + sx$ and $\Delta\Delta' = h^2$ gives the ratios

$$H\left(1 + \frac{s - s'\Delta/h}{2}x\right), \qquad H\left(1 + \frac{s + s'\Delta/h}{2}x\right). \tag{6}$$

For example, $K/(3-\phi) = H$ and $x\sqrt{\Delta} = 1 + x^2$ give these identities immediately after rationalizing $1 + y^2$, where $y^2 - s'y - 1 = 0$. The formal-codegree theorem and independence of $1, \phi, x, \phi x$ therefore force the four restriction columns, in that basis order, to be

$$(1,0,0,0), \quad (1,0,a,b), \quad (1,1,p,q), \quad (1,1,a+b-p,a+2b-q). \tag{7}$$

All entries must be nonnegative integers. In particular, the two occurrences of f lie one in each of the last two columns; this also follows from the constant term $H = 1 + \phi$ in (6). For a fully explicit calculation, put

$$P = a^3 + 2a^2b + 4a - b^3, \qquad Q = a^3 + 3a^2b + ab^2 + 4a - 2b^3 - 4b.$$

These satisfy $Hs'\Delta = P + Q\phi$. If $N = r^2$, define $(d_p, d_q) = (P/r, Q/r)$; if $N = 5r^2$, define

$$(d_p, d_q) = \left(\frac{2Q - P}{5r}, \frac{2P + Q}{5r}\right). \qquad p = \frac{a + b - d_p}{2}, \quad q = \frac{a + 2b - d_q}{2}. \tag{8}$$

The formulas for p, q apply in both cases.

Lemma 3.13. *If $b > 0$ and (4) admits a pivotal categorification, then $x < 36$, and hence $a + b\phi < 36$.*

Proof. Pass to a pseudo-unitary categorification by Lemma 3.12. Let L_X, L_Y denote the known twist-one masses in $I(X), I(Y)$ contributed by the four columns (7). Put

$$z_1 = p + q\phi, \qquad z_2 = (a + b - p) + (a + 2b - q)\phi, \qquad z_1 + z_2 = Hs.$$

The last two central dimensions are $H + z_1 x, H + z_2 x$. Consequently

$$\begin{aligned} L := L_X + \phi L_Y &= sx^2 + H(z_1 + z_2) + x(z_1^2 + z_2^2) \\ &\geq sx^2 + H^2 s + \tfrac{1}{2} H^2 s^2 x =: L_0. \end{aligned}$$

[11, Proposition 3.2, with the weaker two-moment bound], applied to X and Y and added with weights $1, \phi$, gives

$$9L \leq D\big(3(1 + \phi^2)x + 2(1 + \phi)\big) = K(1 + x^2)(3Kx + 2H).$$

Define

$$F(t) = \tfrac{3}{2}(2 - \phi)t^2 - (8\phi + 6)t - \tfrac{9}{2}(3\phi + 6).$$

Substituting $s = x - x^{-1}$ gives the exact identity

$$9L_0 - K(1 + x^2)(3Kx + 2H) = \frac{x^2 + 1}{x} F(x) + \frac{18}{x}.$$

The left-hand side is nonpositive, so $F(x) < 0$. On the other hand,

$$2F(36) = 7290 - 4491\phi > 0, \qquad F'(36) = 210 - 116\phi > 0.$$

For instance $\phi < 34/21$ gives the first strict lower bound $132/7$. Since F has positive leading coefficient, it is positive for all $t \geq 36$. This proves $x < 36$. □

Proof of Theorem 3.11. If $b = 0$, the subring on $1, X$ has $X^2 = 1 + aX$. The rank-two moment obstruction [11, Proposition 5.1] forces $a = 0$ or 1, giving the first two rings in the statement. Suppose $b > 0$. Lemmas 3.12 and 3.13 give an exhaustive finite procedure. The bound $a + b\phi < 36$ restricts the parameters to $0 \leq a \leq 35$, $1 \leq b \leq 22$, and is equivalent in that box to

$$72 - 2a - b > 0, \qquad 5b^2 < (72 - 2a - b)^2.$$

For each such pair, compute u, v, N from (5) and retain it only if N is a square or five times a square. Integer square roots test these alternatives exactly. Next compute the four forced columns in (7) and reject a negative or nonintegral entry. For every remaining pair, compute the masses L_X, L_Y and test the two-moment inequalities. Each operation takes place in the algebra defined by $\phi^2 = 1 + \phi$ and $x^2 = 1 + (a + b\phi)x$.

The complete implementation is `companion_software/verify_unbounded_fibonacci.py`. It exhausts the stated parameter box, constructs the columns and exact moment deficits, and verifies positive deficits by rational bounds. The inequalities $3/2 < \phi < 2$, $x > 0$ and $x < a + b\phi + 1$ supply its sign witnesses; the final bound follows from $x = s + x^{-1}$ and $x > 1$. The generated record includes every surviving parameter and its realization or exclusion certificate. There are no catalogue lookups or parameter-specific calculations in the argument. Thus only $(a, b) = (1, 1)$ remains when $b > 0$. Its fusion rules are those of $\mathrm{Ad}(E_8)$ [3], already identified in the positive-model table as $F_{3,19}$. The three stated rings have the indicated unitary models, completing the proof. □

3.5 Proper fusion subrings and Galois reduction

Proposition 3.14. *Let R be a self-dual rank-four fusion ring. Unless $R \otimes_{\mathbb{Z}} \mathbb{Q} \cong K \times K$ for a real quadratic field K, every pivotal categorification of R has a pseudo-unitary Galois conjugate, or is already pseudo-unitary.*

Proof. The rational algebra is a product of totally real number fields whose degrees sum to four. Let K_0 be the field corresponding to the Frobenius–Perron character and put $D = \mathrm{FPdim}(R)$. If $K_0 = \mathbb{Q}$, the category is integral and pseudo-unitary. If $D \in \mathbb{Q}$, it is weakly integral and again pseudo-unitary. Assume therefore that both are nonrational.

A pivotal categorification is spherical. Its dimension field must contain D. If $[K_0 : \mathbb{Q}] = 4$, every character belongs to the same Galois orbit. If its degree is three, the only other field factor is rational, so cannot supply the spherical dimension character. If its degree is two, then D generates K_0. Every other field factor is rational, or there is one further quadratic factor K_1. Rational factors cannot contain D, and a distinct quadratic field K_1 cannot contain it either. Except in the indicated repeated-field case, the dimension character therefore belongs to the Frobenius–Perron orbit. Galois conjugation of the spherical category gives the claimed pseudo-unitary realization. □

The repeated-field exception cannot be removed for individual realizations: the product of a Fibonacci category and its Yang–Lee Galois conjugate has rank four and categorical global dimension 5, whereas the Frobenius–Perron dimension of its fusion ring is $5\phi^2$, with $\phi = (1 + \sqrt{5})/2$. Its global dimension stays 5 under Galois conjugation. Its fusion ring nevertheless has a unitary realization, namely the product of two unitary Fibonacci categories.

Theorem 3.15. *A rank-four fusion ring with a proper nontrivial fusion subring admits a pivotal categorification if and only if it is one of the thirteen rings in the realization table other than $F_{1,6}$ and $F_{2,15}$. All thirteen admit unitary categorifications. There is no multiplicity bound.*

Proof. If the ring has a nontrivial invertible basis element, Corollary 3.10 gives precisely eleven possibilities. Every such ring has a proper pointed subring (or, in the pointed rank-four cases, a subgroup of order two).

Suppose the pointed part is trivial. The unrestricted non-self-dual classification implies that a pivotal-categorifiable ring of this form is self-dual. A proper nontrivial fusion subcategory has rank two or three and inherits the pivotal structure. In rank two, the moment obstruction [11, Proposition 5.1] leaves the Fibonacci fusion ring. In rank three, the pivotal classification [6] leaves only the ring of $\mathrm{PSU}(2)_5$, because every other positive rank-three ring has a nontrivial invertible object. But the single remaining simple W outside such a rank-three subcategory must satisfy $VW = n_V W$ for every simple V in it: reciprocity excludes summands in the subcategory from VW. Thus $\mathrm{FPdim}(V) = n_V \in \mathbb{Z}$, contradicting the nonintegral dimensions of $\mathrm{PSU}(2)_5$. A Fibonacci subring is therefore present. Theorem 3.11 leaves exactly $\mathrm{Fib} \boxtimes \mathrm{Fib}$ and $\mathrm{Ad}(E_8)$ when the pointed part is trivial. Together with the eleven preceding rings these are the stated thirteen. The displayed unitary models prove sufficiency. □

4 Dimension fields and Galois pseudo-unitarity

4.1 A multiplicity bound for the quadratic character branch

We now remove the multiplicity restriction from another entire rational algebra type. The argument uses only two central twist moments, the orthogonality of rational characters, and the unrestricted exclusion of R_n.

Lemma 4.1 (A codegree moment bound)**.** *Let R be a self-dual rank-four fusion ring with a pseudo-unitary categorification. Let $d_0 = 1, d_1, d_2, d_3$ be its Frobenius–Perron dimensions, $D = \sum_i d_i^2$, and let $f_0, \dots, f_3$ be its formal codegrees, counted with multiplicity. Put*

$$w = \sum_j f_j^{-2}, \qquad S = \sum_i d_i.$$

Then

$$D(1-3w) + (1-w)S + 4w - \frac{2}{D} \geq 0, \qquad S \leq 1 + \sqrt{3(D-1)}. \tag{9}$$

Proof. Apply the global spherical codegree bound of [11, Proposition 6.2] to the canonical spherical structure. Its positive simple dimensions make the absolute-value sum equal to S. The second assertion is Cauchy–Schwarz applied to d_1, d_2, d_3. □

Corollary 4.2 (A common dimension cutoff)**.** *Under the hypotheses of Lemma 4.1, if $w \geq 3/8 - 2/D$, then $D < 196$ and the multiplicity is at most thirteen.*

Proof. Suppose $D \geq 196$. The left side of (9) decreases as a function of w, since its coefficient is $-3D - S + 4 < 0$. Thus

$$\begin{aligned} 0 &\leq D(1-3w) + (1-w)S + 4w - 2/D \\ &\leq -\frac{D}{8} + \frac{15}{2} + \left(\frac{5}{8} + \frac{2}{D}\right) S - \frac{10}{D} \\ &< -\frac{D}{8} + \frac{49}{6} + \frac{7}{6}\sqrt{D} \\ &= -\frac{(\sqrt{D} - 14)(3\sqrt{D} + 14)}{24} \leq 0. \end{aligned}$$

The strict estimate uses $5/8 + 2/D < 2/3$ and $S \leq 1 + \sqrt{3(D-1)} < 1 + (7/4)\sqrt{D}$. This contradiction proves $D < 196$. Every entry of a real symmetric nonnegative fusion matrix is at most its operator norm, its Frobenius–Perron eigenvalue d_i. Since $d_i^2 < D < 196$, every integral structure constant is at most thirteen. □

Lemma 4.3. *Suppose a self-dual rank-four fusion ring has two rational characters with formal codegrees m, n. Then m, n are integers at least two. The pairs $(3,4)$ and $(3,5)$ are impossible. If $m = n = 3$, then the ring is R_k for an integer $k \geq 1$.*

Proof. The character values are rational algebraic integers, hence integers, and the first value of every character is one. Its formal codegree is the sum of the four squared values. Distinct character vectors are orthogonal because the fusion matrices are commuting real symmetric matrices. A rational character of codegree three therefore has nontrivial entries $0, \pm 1, \pm 1$. A character of codegree four has nontrivial entries $\pm 1, \pm 1, \pm 1$, while a character of codegree five has entries $\pm 2, 0, 0$, up to permutation. Its scalar product with a codegree-three character is odd in both cases, and cannot vanish. This excludes the two pairs. A codegree equal to one would give a vector $(1, 0, 0, 0)$, which cannot be orthogonal to the positive Frobenius–Perron character; thus $m, n \geq 2$.

Suppose now $m = n = 3$. Orthogonality to the positive dimension vector, whose entries are all at least one, forces the two nonzero nontrivial entries of each rational character to have opposite signs. Their zero coordinates are distinct: otherwise their scalar product would again be odd. Orthogonality between the two characters then forces a chain of two differences of Frobenius–Perron dimensions equal to one. After ordering the basis along this chain, the characters and dimension vector are

$$(1, 0, 1, -1), \qquad (1, 1, -1, 0), \qquad (1, \delta - 1, \delta, \delta + 1).$$

Apply the two rational characters to the six products of the three nontrivial basis elements, using the symmetric parameter tuple $(a, b, c, d, e, f, g, h, i, j)$. The first character gives

$$c = b + 1, \quad e = d, \quad f = e, \quad h = g, \quad i = h + 1, \quad j = i;$$

the second gives

$$a = b, \quad d = b + 1, \quad e = c, \quad g = d, \quad h = e, \quad i = f + 1.$$

Set $k = d = b + 1 \geq 1$. These linear equations give exactly

$$(a, b, c, d, e, f, g, h, i, j) = (k - 1, k - 1, k, k, k, k, k, k, k + 1, k + 1),$$

which is the defining multiplication of R_k. □

Theorem 4.4. *Let R be a self-dual rank-four fusion ring such that*

$$R \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q} \times \mathbb{Q} \times K$$

for a real quadratic field K. If R admits a pivotal categorification, then its multiplicity is at most thirteen. Consequently the bounded classification certificates through multiplicity thirteen give the unrestricted pivotal classification for this rational algebra type.

Proof. Proposition 3.14 gives a pseudo-unitary Galois conjugate or an already pseudo-unitary category. We may therefore apply Lemma 4.1 to the original Frobenius–Perron dimensions.

First suppose the Frobenius–Perron character is rational. All its values are integers. The formal-codegree field condition then makes all four formal codegrees positive integers. Since their reciprocals sum to one, their maximum is at most 42. Here is the elementary four-term unit-fraction bound: after ordering the denominators $a \leq b \leq c \leq d$, one has $2 \leq a \leq 4$. If $a = 4$, all four are 4. If $a = 3$, then $b = 3$ or 4, and the equation for c, d gives $d \leq 12$. If $a = 2$, then $3 \leq b \leq 6$; for $b = 3, 4, 5, 6$ the maximum possible d is respectively $42, 20, 10, 6$. These follow directly from $1/c + 1/d = 1 - 1/a - 1/b$, with $c \leq d$ and a positive right side. As D is the largest formal codegree, $D \leq 42$.

Suppose now that the Frobenius–Perron character is quadratic. Its two characters have codegrees D, D', where D' is the Galois conjugate of D and is allowed to equal D. The remaining two codegrees m, n come from the rational characters. Put $u = 1/m$, $v = 1/n$ and $c = 1 - u - v$. The reciprocal identity gives $1/D' = c - 1/D$, and therefore

$$w = u^2 + v^2 + (c - 1/D)^2 + D^{-2} = u^2 + v^2 + c^2 - \frac{2c}{D} + \frac{2}{D^2}.$$

We have $0 < c < 1$. If $m = n = 3$, Lemma 4.3 identifies the ring as R_k. The unrestricted quadratic-family theorem excludes every $k \geq 2$; for $k = 1$ one has $D = (39 + 9\sqrt{13})/2 < 196$.

For every other possible pair, we claim

$$u^2 + v^2 + (1 - u - v)^2 \geq \frac{3}{8}.$$

If one denominator is two, complete the square in the other reciprocal:

$$\frac{1}{4} + v^2 + (\tfrac{1}{2} - v)^2 = \frac{3}{8} + 2(v - \tfrac{1}{4})^2.$$

If both denominators are at least four, the expression decreases in each of u, v on $0 < u, v \leq 1/4$, so its minimum occurs at $u = v = 1/4$ and is $3/8$. The only remaining case has one

denominator three. The other is neither three, four nor five, by Lemma 4.3, and the case two was already covered. Thus $v \le 1/6$, and

$$\frac{1}{9} + v^2 + (\tfrac{2}{3} - v)^2 \ge \frac{7}{18} > \frac{3}{8}.$$

This proves the claim and gives $w \ge 3/8 - 2/D$.

Corollary 4.2 now gives $D < 196$, completing all cases.

Finally, each fusion matrix N_i is real symmetric and nonnegative, with spectral radius d_i. Hence each of its entries is at most its operator norm d_i. Since $d_i^2 < D < 196$, every fusion coefficient is less than fourteen, and its integrality gives the claimed multiplicity bound. □

Corollary 4.5. *The self-dual rank-four fusion rings with rational algebra $\mathbb{Q} \times \mathbb{Q} \times K$, for a real quadratic field K, which admit pivotal categorifications are exactly the four rings of*

$$\mathrm{Rep}(D_{10}), \qquad \mathrm{PSU}(2)_6, \qquad 2D2, \qquad \mathcal{H}_1,$$

where D_{10} is the dihedral group of order ten. Each has a unitary categorification. If the ring has no nontrivial invertible basis object, only the ring of $\mathcal{H}_1$ occurs.

Proof. Theorem 4.4 reduces the assertion to the exact classification certificates through multiplicity thirteen. The exact model verifier reads the full tensors in `data/models.json`. For every self-dual model it constructs $A = N_X + 2N_Y + 4N_Z$, checks that its characteristic polynomial is squarefree, and factors it over $\mathbb{Q}$. Thus A generates the rational fusion algebra, whose factor degrees identify its algebra type. The routine `companion_software/verify_unbounded_quadratic_branch.py` verifies that precisely the four stated models have two rational factors and one quadratic factor. This computation uses the model tensors themselves, independently of their census indices. The first three stated rings have a nontrivial invertible basis object; $\mathcal{H}_1$ does not. Their unitary realizations are those cited in the positive-model table. □

Corollary 4.6 (The integral self-dual case)**.** *Every complex-categorifiable self-dual rank-four fusion ring with rational Frobenius–Perron character has* $\mathrm{FPdim}(R) \le 42$ *and multiplicity at most six. The only such rings are those of* $\mathrm{Vec}_{C_2 \times C_2}$ *and* $\mathrm{Rep}(D_{10})$.

Proof. A rational Frobenius–Perron character is integral. Any categorification is therefore pseudo-unitary, so the rational-character argument in the proof of Theorem 4.4 applies without a pivotal hypothesis: the four formal codegrees are positive integers with reciprocal sum one, and their maximum D is at most 42. Every fusion coefficient is at most $\max_i d_i < \sqrt{42} < 7$. The bounded-classification certificates through multiplicity six leave exactly the two stated self-dual integral rings, both with standard unitary realizations. □

4.2 A uniform bound for the repeated-quadratic exception

All norms in the next argument are taken in the specified quadratic field, even when the element being normed is rational.

Theorem 4.7. *Let $\mathcal{C}$ be a self-dual rank-four pivotal fusion category. If no Galois conjugate of $\mathcal{C}$ is pseudo-unitary, then*

$$\mathrm{FPdim}(\mathcal{C}) < 196.$$

Consequently its fusion-rule multiplicity is at most 13.

Proof. Self-duality makes the pivotal structure spherical. Proposition 3.14 reduces the question to $R \otimes \mathbb{Q} \cong K \times K$ for a real quadratic field K. Write $D = \mathrm{FPdim}(\mathcal{C})$ and let D' denote its conjugate in K. Rational D would make $\mathcal{C}$ weakly integral and pseudo-unitary, so D is nonrational. The two remaining characters give formal codegrees E, E', conjugate in K, possibly equal if $E \in \mathbb{Q}$. All four are positive.

If the spherical dimension character belongs to the Frobenius–Perron orbit, its category has a pseudo-unitary Galois conjugate. We may therefore suppose its global dimension belongs to the other orbit, and conjugate the category so that its global dimension is $E \geq E'$. Every character value on a basis element has absolute value at most its Frobenius–Perron dimension; hence every formal codegree is at most D. Thus $E \leq D$, and equality would itself give pseudo-unitarity. Hence $E < D$.

The formal-codegree theorem supplies a central simple of categorical dimension E/D. In particular, E/D is an algebraic integer in K. Set

$$n = \mathrm{Norm}_{K/\mathbb{Q}}(D) = DD', \qquad m = \mathrm{Norm}_{K/\mathbb{Q}}(E) = EE', \qquad k = m/n.$$

The norm of the nonzero algebraic integer E/D is a positive integer, so $k \in \mathbb{Z}_{\geq 1}$. The definitions remain valid when $E = E'$, in which case $m = E^2$. Put

$$A = D + D', \quad B = E + E', \quad p = A/n.$$

The reciprocal-codegree identity, counting both characters when their codegrees coincide, gives

$$\frac{A}{n} + \frac{B}{kn} = 1. \tag{10}$$

Consequently

$$S := \frac{1}{D^2} + \frac{1}{(D')^2} + \frac{1}{E^2} + \frac{1}{(E')^2} = p^2 + (1-p)^2 - \frac{2(1+1/k)}{n}. \tag{11}$$

Ostrik's spherical inequality [5, Theorem 3.1.1] gives

$$S \leq \frac{1}{2} + \frac{1}{2E}. \tag{12}$$

For completeness, the squared central dimensions give a positive measure on twists with total mass E^2 and first moment E. The four twist-one constituents of $I(1)$ have squared mass $h = E^2 S$. The remaining mass and first moment are $E^2 - h$ and $E - h$, so $h - E \leq |E - h| \leq E^2 - h$. This proves the displayed inequality without any pseudo-unitarity hypothesis. Every formal codegree of a nontrivial self-dual fusion ring is strictly greater than one: it is $1 + \sum_{X \neq 1} \chi(X)^2$, and the character cannot vanish on every nonunit basis element because $[X^2 : 1] = 1$. Thus $D' > 1$, so $n > D$; furthermore $E \geq \sqrt{kn} > \sqrt{D}$.

Suppose for a contradiction that $D \geq 196$. First let $k \geq 2$. The function $t + kn/t$ is increasing for $t \geq \sqrt{kn}$, so $E \leq D$ implies

$$B \leq D + kn/D = A + (k-1)n/D.$$

Using (10) we obtain

$$p \geq \frac{k}{k+1} - \frac{k-1}{(k+1)D} \geq \frac{2}{3} - \frac{1}{3D} > \frac{1}{2}.$$

The last middle expression increases with k because $D > 2$. Equation (11), $n > D$, and $k \geq 2$ now give

$$\begin{aligned} S - \tfrac{1}{2} &> 2\left(\frac{1}{6} - \frac{1}{3D}\right)^2 - \frac{3}{D} \\ &\geq 2\left(\frac{1}{6} - \frac{1}{588}\right)^2 - \frac{3}{196} = \frac{1}{28} + \frac{589}{172872} > \frac{1}{28}. \end{aligned}$$

The expression in the first line is increasing for $D \geq 196$. Since $E > \sqrt{D} \geq 14$, this contradicts (12).

It remains to treat $k = 1$. The algebraic integer E/D is now a totally positive unit of norm one. Put $r = D/E > 1$. Its two conjugates are r, r^{-1}, so $r + r^{-1}$ is an integer strictly greater than two. Therefore

$$r \geq \frac{3+\sqrt{5}}{2} > \frac{5}{2}.$$

Also $E \geq \sqrt{n}$ implies $r < \sqrt{D}$. Since $E = D/r$ and $E' = rD'$, the reciprocal identity gives

$$p = \frac{r}{r+1} - \frac{r-1}{D}.$$

For fixed D the right-hand side is a concave function of r. Its minimum on $[5/2, \sqrt{D}]$ occurs at an endpoint. At the first endpoint it equals $5/7 - 3/(2D)$, whereas at the second it is

$$1 - \frac{1}{\sqrt{D}+1} - \frac{1}{\sqrt{D}} + \frac{1}{D} > 1 - \frac{2}{\sqrt{D}} \geq \frac{6}{7}.$$

Thus $p \geq 5/7 - 3/(2D) > 1/2$. Again (11) gives

$$\begin{aligned} S - \tfrac{1}{2} &> 2\left(\frac{3}{14} - \frac{3}{2D}\right)^2 - \frac{4}{D} \\ &\geq 2\left(\frac{3}{14} - \frac{3}{392}\right)^2 - \frac{4}{196} = \frac{1}{28} + \frac{2249}{76832} > \frac{1}{28}, \end{aligned}$$

again contradicting (12) and $E > 14$. This proves $D < 196$ in both cases.

Finally every fusion matrix is real symmetric, and its operator norm is its Frobenius–Perron eigenvalue d_X. Thus $N_{XY}^Z \leq d_X < \sqrt{D} < 14$ for every structure constant. The multiplicity is an integer and is therefore at most 13. □

Corollary 4.8. *A rank-four fusion ring admits a pivotal categorification over $\mathbb{C}$ if and only if it admits a pseudo-unitary categorification. No multiplicity bound is imposed in this statement.*

Proof. For non-self-dual rings use the unrestricted non-self-dual theorem. For a self-dual pivotal category with a pseudo-unitary Galois conjugate there is nothing to prove. Otherwise Theorem 4.7 puts its fusion ring inside the verified bounded classification through multiplicity thirteen. Every positive pivotal ring in that classification has a unitary, and hence pseudo-unitary, realization. Conversely, a pseudo-unitary category has its canonical pivotal structure. □

Corollary 4.9. *Every pivotal rank-four category whose fusion ring has no proper nontrivial fusion subring has a pseudo-unitary Galois conjugate.*

Proof. Such a categorification is self-dual by the unrestricted non-self-dual classification. A hypothetical exception to Galois pseudo-unitarity would have multiplicity at most 13 and rational fusion algebra $K \times K$. The bounded classification leaves only the rings of $\mathrm{PSU}(2)_7$ and $\mathcal{H}_1$ in the absence of a proper nontrivial fusion subring. Their rational fusion algebras have degree patterns $1 + 3$ and $1 + 1 + 2$, respectively, rather than $2 + 2$. This is a contradiction. □

Remark 4.10. Theorem 4.7 bounds the case where the given category has no pseudo-unitary Galois conjugate. The separate moment argument of Theorem 4.13 bounds the pseudo-unitary case for a product of two quadratic fields as well. Thus this repeated-field possibility creates no exception in the unrestricted classification by rational algebra type. The exception for individual realizations is real: the product of a Fibonacci category and its Yang–Lee conjugate has global dimension $E = 5$, $D = 5\phi^2$, and $n = m = 25$, so none of its Galois conjugates is pseudo-unitary even though its fusion ring has a unitary realization.

4.3 Products of two quadratic character fields

The preceding moment bound also treats both quadratic factors at once. The fields in the next theorem are allowed to coincide.

Lemma 4.11. *Let R be a self-dual fusion ring. A character whose value field is quadratic cannot have formal codegree three if its conjugate also has formal codegree three.*

Proof. First, a real quadratic algebraic integer α for which both conjugates have absolute value at most $\sqrt{2}$ belongs to

$$\{0, 1, -1, \sqrt{2}, -\sqrt{2}\}.$$

For a rational α this is immediate. Otherwise write its minimal polynomial as $t^2 - Tt + N$. Its trace satisfies $|T| \leq 2$. If $T = 0$, then $-N$ is a positive nonsquare integer at most two, so $\alpha = \pm\sqrt{2}$. If $|T| = 1$, irreducibility and real roots force $N \leq -1$; one root then has absolute value at least $(1 + \sqrt{5})/2 > \sqrt{2}$. If $|T| = 2$, they force $N \leq 0$, and one root has absolute value at least two. These contradictions prove the claim.

Suppose χ and its quadratic conjugate χ' both have codegree three. Since the ring is self-dual, the codegree is the sum of the squares of the real character values. The unit contributes one, so each nonunit value and its conjugate have absolute value at most $\sqrt{2}$. Some value is irrational because the character's value field is quadratic. The preceding claim makes it $\pm\sqrt{2}$, and the remaining nonunit values must vanish. Thus the two character vectors, in a suitable common ordering of the basis, are

$$(1, \sqrt{2}, 0, \ldots, 0), \qquad (1, -\sqrt{2}, 0, \ldots, 0).$$

Their scalar product is -1, contradicting the orthogonality of distinct characters of a self-dual fusion ring. □

Lemma 4.12. *Let E, E' be the formal codegrees of a quadratic character orbit of a complex-categorifiable self-dual rank-four fusion ring. Then $EE' \geq 16$.*

Proof. Set $m = EE'$ and $B = E + E'$. Both are positive integers. The formal codegrees are totally positive, so $B^2 \geq 4m$. Their reciprocal sum is strictly less than one, since there are two other characters and all codegrees are positive. Hence $B < m$.

The d-number criterion gives $m \mid B^2$. If E is nonrational, apply [11, Proposition 2.1] to $t^2 - Bt + m$. If E is rational, then $E = E'$ is an integer and the same divisibility follows from $B = 2E$ and $m = E^2$.

Here is a short exact small-norm argument. Write $m = ab^2$, where a is a positive squarefree integer and b is a positive integer. The divisibility $m \mid B^2$ implies $B = abt$ for a positive integer t. The inequalities above become

$$t < b, \qquad at^2 \geq 4.$$

If $m < 16$, then $b \leq 3$. The case $b = 1$ is impossible. If $b = 2$, then $a \leq 3$ and $t = 1$, contradicting $at^2 \geq 4$. If $b = 3$, then $a = 1$ and the only possibility is $t = 2$. Thus the unique remaining pair is $(m, B) = (9, 6)$, which gives $E = E' = 3$. Lemma 4.11 excludes this pair. □

Theorem 4.13. *Let R be a self-dual rank-four fusion ring with*

$$R \otimes_{\mathbb{Z}} \mathbb{Q} \cong K_1 \times K_2,$$

where K_1, K_2 are real quadratic fields, possibly equal. If R admits a pivotal categorification, then

$$\mathrm{FPdim}(R) < 196, \qquad \mu(R) \leq 13.$$

The only pivotal-categorifiable rings of this algebra type are $\mathrm{Fib} \boxtimes \mathrm{Vec}_{\mathbb{Z}/2}$ *and* $\mathrm{Fib} \boxtimes \mathrm{Fib}$*, and both admit unitary categorifications.*

Proof. By Corollary 4.8, a pseudo-unitary categorification exists. Number the characters so that D, D' are the codegrees of the Frobenius–Perron orbit and E, E' are the codegrees of the other quadratic orbit. Conjugate codegrees are allowed to coincide. Put

$$m = EE', \qquad B = E + E', \qquad u = B/m, \qquad c = 1 - u.$$

The reciprocal-codegree identity gives $1/D' = c - 1/D$, with $0 < c < 1$. Consequently the quantity in Lemma 4.1 satisfies

$$\begin{aligned} w &= \frac{1}{D^2} + \frac{1}{(D')^2} + \frac{1}{E^2} + \frac{1}{(E')^2} \\ &= u^2 + (1-u)^2 - \frac{2}{m} - \frac{2c}{D} + \frac{2}{D^2} \\ &\geq \frac{1}{2} - \frac{2}{m} - \frac{2}{D} \geq \frac{3}{8} - \frac{2}{D}, \end{aligned}$$

where the last inequality is Lemma 4.12.

Corollary 4.2 gives $D < 196$ and multiplicity at most thirteen.

The bounded-classification certificates through multiplicity thirteen now apply. Of the positive self-dual rings, precisely $F_{1,2}$ and $F_{1,5}$ have a rational algebra consisting of two quadratic factors. They are the two rings in the statement. For the first, the algebra is the tensor product of the Fibonacci quadratic field with $\mathbb{Q}[\mathbb{Z}/2] \cong \mathbb{Q} \times \mathbb{Q}$; for the second, it is $\mathbb{Q}(\sqrt{5}) \otimes_{\mathbb{Q}} \mathbb{Q}(\sqrt{5})$. Both are $\mathbb{Q}(\sqrt{5}) \times \mathbb{Q}(\sqrt{5})$ and have the unitary models given in the realization table. □

Corollary 4.14. *No self-dual rank-four fusion ring with rational algebra* $K_1 \times K_2$*, for distinct real quadratic fields* K_1, K_2*, admits a pivotal categorification. No multiplicity restriction is required.*

4.4 The cubic branch reduces to rational codegree three or four

Proposition 4.15. *Let* R *be a self-dual rank-four fusion ring with* $R_{\mathbb{Q}} \cong \mathbb{Q} \times K$*, where* K *is a totally real cubic field. Let* m *be the formal codegree of the rational character. If* R *admits a pivotal categorification, then* $m = 3$ *or* $m = 4$*. More precisely,* $m = 1, 5$ *are impossible already at the ring level, while* $m = 2$ *or* $m \geq 6$ *forces* $D = \mathrm{FPdim}(R) < 196$ *and multiplicity at most thirteen, and is excluded by the bounded classification.*

Proof. The rational character has integer values, with first value one, so its formal codegree m is a positive integer. If $m = 1$, its vector is $(1, 0, 0, 0)$, which cannot be orthogonal to the positive Frobenius–Perron character. If $m = 5$, its nontrivial entries have squared sum four, so the vector is $(1, \pm 2, 0, 0)$ up to permutation. Orthogonality to the positive dimension vector would require $1 \pm 2d_i = 0$, contrary to $d_i \geq 1$.

By Proposition 3.14, a pivotal categorification has a pseudo-unitary Galois conjugate or is already pseudo-unitary. If its Frobenius–Perron character is rational, Corollary 4.6 already excludes the present algebra type. We may therefore assume that it belongs to the cubic orbit. The formal codegrees are then m, D, D_2, D_3, counted with their character multiplicities. The identity for their reciprocals gives

$$D_2^{-1} + D_3^{-1} = 1 - \frac{1}{m} - \frac{1}{D}.$$

Writing w as in Lemma 4.1, Cauchy–Schwarz therefore gives

$$\begin{aligned} w &\geq \frac{1}{m^2} + \frac{1}{2}\left(1 - \frac{1}{m} - \frac{1}{D}\right)^2 + \frac{1}{D^2} \\ &= \left[\frac{1}{m^2} + \frac{1}{2}\left(1 - \frac{1}{m}\right)^2\right] - \frac{1 - 1/m}{D} + \frac{3}{2D^2}. \end{aligned}$$

For $m = 2$, the bracket is $3/8$. For $m \geq 6$, put $u = 1/m \leq 1/6$; the function $u^2 + (1-u)^2/2$ decreases on $[0, 1/6]$ and equals $3/8$ at $u = 1/6$. Thus in both cases

$$w \geq \frac{3}{8} - \frac{1}{D} \geq \frac{3}{8} - \frac{2}{D}.$$

Corollary 4.2 now gives $D < 196$, hence multiplicity at most thirteen. The bounded-classification census has only one positive self-dual ring with rational algebra $\mathbb{Q} \times$ (cubic field), that of $\mathrm{PSU}(2)_7$, whose rational formal codegree is three. It therefore excludes every case $m = 2$ or $m \geq 6$ under consideration. Only $m = 3, 4$ remain for the unrestricted investigation. □

5 A bounded induction difference in the cubic codegree-three branch

Consider a self-dual pseudo-unitary rank-four category whose rational Grothendieck algebra is $\mathbb{Q} \times K$, with K cyclic cubic, and whose rational character has formal codegree three. Normalize this character to $\rho = (1, 0, 1, -1)$. In the resulting basis $(1, X, Y, Z)$ the tensor tuple has the form

$$(a, b, b+1, d, d, d, g, g, g+1, g+1), \qquad ad + (2b+1)g - b^2 - 2d^2 + 1 = 0.$$

Let x, y, z be the Frobenius–Perron dimensions. Character orthogonality gives $z = y + 1$. The cubic character hypothesis gives the unique rational relation $(1, 0, 1, -1)$ among $1, x, y, z$.

Write $D = \mathrm{FPdim}\,\mathcal{C}$ and $I(1) = 1 \oplus B \oplus A \oplus A'$, where B corresponds to ρ. Its dimension is $D/3$. Comparison of its restriction dimension in the cubic field forces

$$F(B) = (1, p, q, q+1), \qquad p = \frac{a + 2d}{3}, \quad q = \frac{b + 2g}{3}, \qquad p, q \in \mathbb{Z}_{\geq 0}.$$

In particular the indicated congruences modulo three are necessary.

Proposition 5.1. *There is an integer $k \in \{-1, 0, 1\}$ such that, on the two restrictions $F(A), F(A')$, the multiplicity differences $[Y] - [Z]$ are $k, -k$. For each remaining central simple W, put*

$$h_W = [F(W) : Z] - [F(W) : Y].$$

Then

$$\sum_{W \notin \{1, B, A, A'\}} h_W^2 = 6 - 2k^2.$$

Thus at most six remaining central simples can have different Y- and Z-multiplicities, independently of the fusion coefficients. If $k = 0$, the multiset of nonzero $|h_W|$ is either six ones or $(2, 1, 1)$. If $|k| = 1$, it is either four ones or (2).

Proof. Let M be the full restriction matrix of central induction, and $Q = \sum_U N_U^2 = MM^{\mathsf{T}}$. The first column of Q is

$$(4, 3p, 3q + 1, 3q + 2)^{\mathsf{T}}.$$

Since precisely the four constituents of $I(1)$ have unit coefficient one in their restrictions, the restriction columns of A, A' sum to $(2, 2p, 2q+1, 2q+1)^{\mathsf{T}}$. Their $Y - Z$ differences therefore sum to zero; denote them by $k, -k$.

Put $v = (1, 0, 1, -1)^{\mathsf{T}}$ and $w = (0, 0, -1, 1)^{\mathsf{T}} = e_0 - v$. Since ρ has codegree three, $Qv = 3v$ and $v^{\mathsf{T}}v = 3$. As $Q_{00} = 4$, we obtain

$$w^{\mathsf{T}}Qw = 4 - 2 \cdot 3 + 3 \cdot 3 = 7.$$

The four known columns contribute $0, 1, k^2, k^2$ to this Gram expression. Subtracting proves the displayed identity. Its nonnegative integer left side forces $k^2 \leq 3$, hence $k \in \{-1, 0, 1\}$. The final assertions are the elementary decompositions of four or six as sums of nonzero integer squares. □

There are corresponding moment identities involving at most six twists. Put $\alpha = \dim A$, $\beta = \dim A'$ and $a_W = \dim W > 0$. Then

$$\begin{aligned}
\sum_W h_W a_W &= \frac{2D}{3} + k(\alpha - \beta), \\
\sum_W h_W a_W \theta_W &= -\frac{D}{3} + k(\alpha - \beta), \\
\sum_W h_W a_W \theta_W^2 &= D\big(\nu_2(Z) - \nu_2(Y)\big) - \frac{D}{3} + k(\alpha - \beta),
\end{aligned}$$

where the sums range only over the remaining central simples. Indeed $\dim I(Z) - \dim I(Y) = D$ since $z - y = 1$, while the known twist-one contribution is $D/3 - k(\alpha - \beta)$. Subtraction from the first and second indicator identities proves the other two formulas. More generally the last right side is valid with ν_n and θ^n for every $n \geq 1$.

Corollary 5.2. *If $|k| = 1$, exactly four remaining central simples have nonzero $Y - Z$ multiplicity difference, and each difference has absolute value one.*

Proof. It remains to exclude the single-term pattern (2). In that case the zeroth and first moments above are real numbers of the same absolute value, since the twist has absolute value one. They differ by D, so they must be negatives of each other. Therefore

$$k(\alpha - \beta) = -D/6.$$

Since $\alpha + \beta = 2D/3 - 1$, one of α, β equals $D/4 - 1/2$. The two dimensions are $D/\sigma(D)$ for the two nontrivial automorphisms σ of the cyclic cubic field. Consequently one such automorphism satisfies

$$\sigma(1/D) = \frac{1}{4} - \frac{1}{2D}.$$

Iterating this affine identity three times gives $1/D = 1/6$, impossible because D has degree three. For completeness, D cannot be rational: otherwise the three cubic characters would all have the same formal codegree D, and the reciprocal-codegree identity would give $1/3 + 3/D = 1$, so $D = 9/2$, contradicting algebraic integrality. □

What this reduction does not prove. The coefficients h_W may be negative, and the positive dimensions a_W are not bounded by the number of nonzero terms. Thus these identities are not sums of a bounded number of roots of unity with unit weights. Neither the positive-measure inequalities nor a cyclotomic length bound can be applied to this signed measure without an additional argument. The reduction bounds the number of central objects that must be analyzed; it does not yet bound the original fusion coefficients or classify this cubic branch. No unrestricted categorification classification is inferred from it.

6 The unrestricted classification problem

6.1 The remaining unrestricted problem

All cases with Frobenius–Perron character field of degree at most two are now classified. The proper-subring theorem also handles arbitrary character degree when a proper nontrivial fusion subring is present. The known rings covered by these two results are the fourteen entries of the realization table other than $F_{1,6}$, the ring of $\mathrm{PSU}(2)_7$.

Question 6.1. Let R be a self-dual rank-four fusion ring with no proper nontrivial fusion subring and with Frobenius–Perron character field of degree three or four. If R admits a pivotal categorification, must it be the ring of $\mathrm{PSU}(2)_7$?

No complete answer is obtained here. A pivotal realization of a ring in this question has a pseudo-unitary Galois conjugate, by Corollary 4.9. Cyclotomicity makes its character field cyclic cubic, cyclic quartic or biquadratic. In the cubic case, Proposition 4.15 further restricts the rational character to formal codegree three or four. The finite classification constrains any additional positive ring to lie outside its certified range, but supplies no upper bound for these remaining field types.

Section 5 gives a further uniform reduction when the rational character in the cubic case has codegree three. At most six central simples distinguish two specified basis objects, and one possible pattern is excluded by Galois symmetry. Their dimensions remain unbounded. In particular, a bound on the number of terms in a signed twist identity is not a bound on its cyclotomic length. These constraints do not settle the remaining question, and no unrestricted classification of all rank-four pivotal rings is claimed.

6.2 All fusion rings versus categorifiable fusion rings

There are two distinct ways an unrestricted classification could be completed. One could parametrize all rank-four fusion rings and then prove a uniform realization-or-exclusion theorem for those parameters. Alternatively one could bound the multiplicity or Frobenius–Perron dimension of every potentially categorifiable ring and use the bounded classification. The latter route does not require classifying all noncategorifiable fusion rings.

Proposition 6.2 (An infinite family of raw fusion rings)**.** *For all integers $a, b \geq 0$, the tensor product of the rank-two based rings $x^2 = 1 + ax$ and $y^2 = 1 + by$ is a self-dual rank-four fusion ring. With $z = xy$, its remaining products are*

$$xz = y + az, \qquad yz = x + bz, \qquad z^2 = 1 + ax + by + abz.$$

Its multiplicity is $\max\{1, a, b, ab\}$*. It admits a complex fusion categorification if and only if* $a, b \in \{0, 1\}$*.*

Proof. The tensor product basis is $1, x, y, xy$; positivity, the unit coefficient, duality and associativity are inherited from the two rank-two based rings. The displayed products give the multiplicity. A categorification would categorify both rank-two subrings, forcing $a, b \in \{0, 1\}$ by [11, Proposition 5.1]. Conversely their pointed or Fibonacci unitary models have the required Deligne products. □

In particular, raw rank-four fusion rings have unbounded multiplicity and cannot be classified by a finite list. This is not an obstruction to a parametric classification: the entire infinite family above is decided by one rank-two theorem.

6.3 Diophantine presentations and finite census data

Appendix B of [11] recalls the self-dual ten-parameter and non-self-dual six-parameter tensor presentations and their necessary and sufficient associativity equations. The bounded classification [11] uses pinned complete repository data through multiplicity 128, comprising 889530 rings. These bounded lists do not parametrize the unbounded nonnegative solutions in disjoint families.

For the following calculation we use the non-self-dual parameters (a, b, c, d, e, f) of that presentation. A useful further elimination illustrates the arithmetic structure. In the non-self-dual case set $u = c - d$ and $P(u) = u^2 - au - 1$. Subtraction of the first two equations and use of the third give

$$b(e - f) = P(u), \qquad dP(u) = -b^2 u.$$

If $b \neq 0$, then $P(u) \neq 0$: an integer root of $u^2 - au - 1$ would have $u = \pm 1$, and $dP(u) = -b^2 u$ would force $b = 0$. Since $P(u) \equiv -1 \pmod{u}$, one has $\gcd(P(u), u) = 1$; consequently

$$P(u) \mid b^2 \qquad (b \neq 0).$$

The branch $b = 0$ is retained separately rather than discarded by division. This divisor restriction can replace an unstructured search by arithmetic branches; it is not, by itself, a bound on the parameters.

6.4 What remains to be proved

A parametric classification of all rank-four fusion rings is a concrete Diophantine problem with a fixed small system of equations, rather than an unformulated search. The proper-subring families and the dimension-field reductions proved here make a systematic approach plausible. Nevertheless, the equations have unbounded integral solutions, and the present arguments do not prove a descent to finitely many seed families in the residual self-dual case.

For the Grothendieck-ring problem the more focused target is the residual cubic or quartic branch of Question 6.1. A uniform bound on its potentially categorifiable parameters, or a uniform moment or arithmetic obstruction for every additional family, would close the pivotal classification. General categorification requires, in addition, control of the possible proper signed sphericalizations. The finite certificate records the outcome of that control over the bounded census; its uniform completeness remains a separate question.

Thus success of the fixed criterion suite at every tested multiplicity is meaningful evidence for further investigation, but neither a proof that all necessary criteria have been found nor a reason to identify raw fusion ring enumeration as the sole remaining obstacle. The companion results supply reductions toward both tasks while keeping their logical roles separate.

The finite paper formulates the unrestricted fifteen-ring classification and the stronger sufficiency of its certificate criteria as conjectures. The reductions proved here support those conjectures but leave Question 6.1 open.

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References

[1] C. Edie-Michell, M. Izumi and D. Penneys, *Classification of $\mathbb{Z}/2\mathbb{Z}$-quadratic unitary fusion categories*, with an appendix by R. Johnson, S.-H. Ng, D. Penneys, J. Roat, M. Titsworth and H. Tucker, Quantum Topol. **16** (2025), 493–568; arXiv:2108.01564v2.

[2] P. Etingof, D. Nikshych and V. Ostrik, *On fusion categories*, Ann. of Math. (2) **162** (2005), 581–642.

[3] D. E. Evans and C. Jones, *Quantum symmetries of noncommutative tori*, arXiv:2404.14466v3.

[4] H. K. Larson, *Pseudo-unitary non-self-dual fusion categories of rank 4*, J. Algebra **415** (2014), 184–213; arXiv:1401.1879.

[5] V. Ostrik, *Remarks on global dimensions of fusion categories*, Contemp. Math. **728** (2019), 169–180; arXiv:1804.08761.

[6] V. Ostrik, *Pivotal fusion categories of rank 3*, Mosc. Math. J. **15** (2015), 373–396, 405; arXiv:1309.4822.

[7] A. Schopieray, *Categorification of integral group rings extended by one dimension*, J. Lond. Math. Soc. (2) **108** (2023), 1617–1641; doi:10.1112/jlms.12789.

[8] A. Bruguières and A. Virelizier, *On the center of fusion categories*, Pacific J. Math. **264** (2013), 1–30; doi:10.2140/pjm.2013.264.1.

[9] A. Schopieray, *Fixed-point-free fusion automorphisms*, in *Quantum Symmetries: Tensor Categories, TQFTs, and Vertex Algebras*, Contemp. Math. **813** (2025), 151–174; doi:10.1090/conm/813/16287; arXiv:2306.01666.

[10] J. Dong, *On Grothedieck rings of rank* 4 *self-dual fusion categories*, arXiv:2505.19153 (2025).

[11] *Rank-four Grothendieck rings through multiplicity 128*, accompanying finite-classification manuscript and exact certificate package, 2026.